



Introduction to Scientific Computing

Graduate Program in Civil Engineering
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Instructors: DSc. Antonio Aquino
DSc. Osvaldo D. Quintana-Ruiz

List of Exercises N° 1

Instructions: All solutions must be implemented in a scientific programming language (*Fortran*, *MATLAB*, or *Python*), with code sections properly commented and explanations provided for the key stages of the implementation. In the text, plain italic letters $a, b, \dots, \alpha, \beta, \dots, A, B, \dots$ denote scalar quantities, whereas boldface italic letters $\mathbf{a}, \mathbf{b}, \dots, \mathbf{A}, \mathbf{B}, \dots$ denote vectors or matrices in a three-dimensional Euclidean space.

1. Gauss elimination with partial pivoting.

Use Gauss elimination *with partial pivoting* to solve $\mathbf{Ax} = \mathbf{b}$ for each of the following cases. Report the solution vector and the number of row interchanges performed.

Case (a):

Case (b):

$$\mathbf{A} = \begin{bmatrix} 3 & -1 & 2 \\ -1 & 4 & -3 \\ 2 & -3 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7 \\ -10 \\ 15 \end{bmatrix} \qquad \mathbf{A} = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 4 & -1 \\ 1 & -1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 20 \\ 8 \\ 11 \end{bmatrix}$$

2. Gauss elimination — singular and rank-deficient systems.

Solve $\mathbf{Ax} = \mathbf{b}$ by Gauss elimination, where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 3 & 2 & -1 \\ 2 & 4 & 0 & -1 & 5 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

3. LU decomposition — reading data from file.

Given the systems below, find the LU factorization $\mathbf{A} = \mathbf{LU}$ and use forward/back substitution to obtain the solution. The coefficient matrix and right-hand side must be read from an external file. Print $\mathbf{L}$, $\mathbf{U}$, and the solution vector $\mathbf{x}$.

System (a):

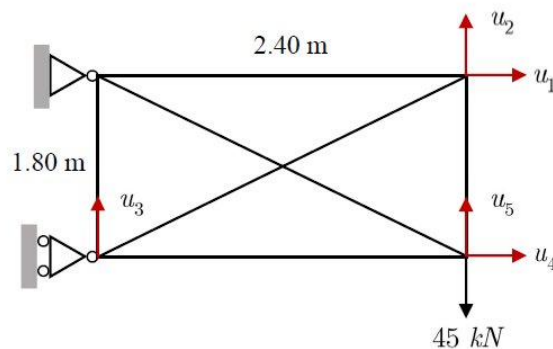
$$\begin{aligned} 3x_2 + x_4 &= 4, \\ 3x_1 + 2x_2 + 4x_3 + x_4 &= 1, \\ 5x_1 - 2x_2 + x_3 &= -6, \\ 7x_1 + x_2 - 5x_3 - 4x_4 &= 7. \end{aligned}$$

System (b):

$$\begin{aligned} x_1 + 3x_2 + 5x_3 + x_4 &= 24, \\ 3x_1 + 9x_2 + 7x_3 + 4x_4 &= 58, \\ 4x_1 + 11x_2 + 9x_3 + 8x_4 &= 83, \\ 5x_1 + 13x_2 + 11x_3 + 7x_4 &= 90. \end{aligned}$$

4. Symmetric system — statically indeterminate truss.

For the statically indeterminate truss shown, the displacement formulation yields the symmetric equations $\mathbf{K}\mathbf{u} = \mathbf{p}$, where



$$\mathbf{K} = 1 \times 10^3 \begin{bmatrix} 27.58 & 7.004 & -7.004 & 0 & 0 \\ 7.004 & 29.57 & -5.253 & 0 & -24.32 \\ -7.004 & -5.253 & 29.57 & 0 & 0 \\ 0 & 0 & 0 & 27.58 & -7.004 \\ 0 & -24.32 & 0 & -7.004 & 29.57 \end{bmatrix} \text{ kN/m,}$$

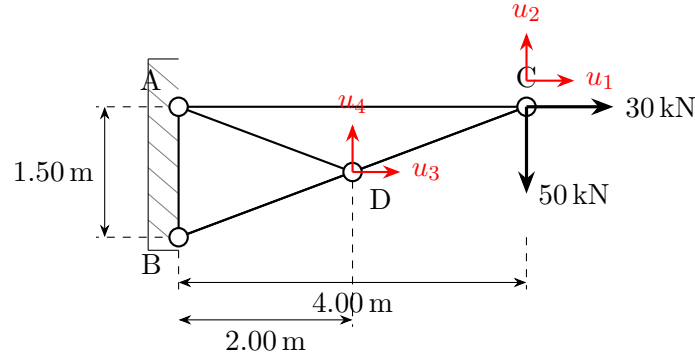
$$\mathbf{p} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -45 \end{bmatrix} \text{ kN.}$$

Determine the joint displacements u_i and analyze which type of solution method is more convenient for this symmetric positive-definite system.

5. Symmetric system — plane truss with combined loading.

The plane truss shown below is fixed to a wall at nodes A (top) and B (bottom) on the left side. Nodes C and D are free. A horizontal load of 30 kN and a vertical downward

load of 50 kN are applied at node C. The free degrees of freedom are u_1 (horizontal at C), u_2 (vertical at C), u_3 (horizontal at D), and u_4 (vertical at D).



All members have the same axial stiffness $EA = 5 \times 10^4$ kN. The global stiffness matrix and load vector assembled for the free degrees of freedom are:

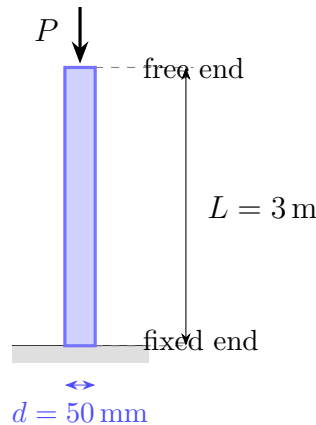
$$\mathbf{K} = 1 \times 10^3 \begin{bmatrix} 35.42 & 4.688 & -12.50 & 0 \\ 4.688 & 22.27 & 0 & -8.333 \\ -12.50 & 0 & 24.31 & 3.906 \\ 0 & -8.333 & 3.906 & 18.75 \end{bmatrix} \text{ kN/m},$$

$$\mathbf{p} = \begin{bmatrix} 30 \\ -50 \\ 0 \\ 0 \end{bmatrix} \text{ kN}.$$

Solve the system using LU decomposition and determine the joint displacements u_1, u_2, u_3, u_4 . Compare the result with the solution obtained by Gauss elimination with partial pivoting and report the residual $\|\mathbf{p} - \mathbf{K}\mathbf{u}\|$ for each method.

6. Bisection method — critical buckling load of a column.

A solid circular steel column is fixed at the base and free at the top (cantilever configuration), as shown in the figure. Its properties are:



The column has modulus of elasticity $E = 200$ GPa and a solid circular cross-section of diameter $d = 50$ mm, giving a moment of inertia $I = \pi d^4/64$.

Applying the boundary conditions of this support configuration to the differential equation of the deflected column leads to the transcendental equation:

$$f(\lambda) = \cos(\lambda) = 0, \quad \lambda = L\sqrt{\frac{P}{EI}}$$

whose first positive root λ^* gives the critical buckling load as:

$$P_{cr} = \frac{EI (\lambda^*)^2}{L^2}$$

- (a) Apply the bisection method to find the first root of $f(\lambda) = \cos(\lambda)$ on the interval $[1.0, 2.0]$ with tolerance $\varepsilon = 10^{-5}$. Report the iteration table showing a , b , midpoint c , $f(c)$, and the interval width $|b - a|$ at each step.
- (b) From the root λ^* obtained, compute the numerical critical load $P_{cr}^{(\text{num})}$.
- (c) Compare the numerical result with the closed-form Euler formula for a fixed-free column:

$$P_{cr}^{(\text{anal})} = \frac{\pi^2 EI}{4L^2}$$

and report the relative error between both results.